Hsuan-Fu (Jeff) Hsu

Embedded Systems and AI Engineering | C++, Python, Raspberry Pi, Arduino, AWS

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EDUCATION

University of Southern California — Los Angeles, CA
Master of Science in Computer Science | GPA: 3.6/4.0 — Jan 2026 - May 2027 (Expected)

National Taipei University — New Taipei City, Taiwan
Bachelor of Science in Computer Science | GPA: 3.5/4.0 — Sept 2020 - June 2024

SKILLS

Languages: C++, Python, C#, TypeScript, SQL

Embedded and Graphics: Raspberry Pi 4, Arduino Nano 33 BLE Sense, I2C, BLE, OpenGL, GLSL

ML and Data: PyTorch, TensorFlow, Sentence Transformers, NumPy, tsfresh

Cloud and Tools: AWS (Lambda, SQS, DynamoDB, CDK), Docker, Linux, Git, pytest, GitHub Actions

EXPERIENCE

National Taipei University — New Taipei City, Taiwan
Undergraduate Researcher | Advisor: Prof. Bor-Shing Lin — July 2022 - Feb 2024

- Co-built a portable NIR sensing device with 2 teammates: **Arduino Nano 33 BLE Sense**, an AS7263 sensor over I2C, and BLE data transfer.
- Built the **Python** signal pipeline for 30 participants at 2 wavelengths; the team's best model reached 10.3% MAPE under leave-one-subject-out testing.
- Won individual NSTC research funding; 4th place in NTPU's project competition (team of 3).

Tech-Management Workshop Co., Ltd. — Taichung, Taiwan
Software Engineer Intern — July 2026 - Aug 2026

- Built a government birth-grant app on my own with **ASP.NET Core MVC** and **SQL Server**, deployed to 29 offices.
- Replaced manual full-dataset uploads with sync of changed records only across the 29 offices.
- Tuned SQL Server processing and storage, cutting database latency by at least 50% and resource use by at least 30%.

PROJECTS

Classroom AC Control System — USC IEEE HackIoT (team of 4) — Oct 2026
Raspberry Pi 4, Python, YOLO, DHT11, Supabase, TypeScript | GitHub

- Set up a **Raspberry Pi 4** with teammates to count people with YOLO at about 7 FPS and read a DHT11 sensor twice a minute, sending data to Supabase.
- Ported a teammate's Python AC model to **TypeScript** (matched over 1,344 hours) and built a Next.js simulator; one room used an estimated 27% less energy in an 8-week simulation.

3D Rendering Engine — Sept 2023 - Dec 2023
C++, OpenGL, GLSL | GitHub

- Wrote a **C++** OBJ/MTL parser and **OpenGL** pipeline with VAO/VBO buffers and model, view, and projection transforms.
- Added texture mapping, multi-light Phong shading, and skybox and environment mapping in **GLSL**.

CatalogBridge — AWS CSV Import Service — Oct 2026
Python, AWS CDK, Lambda, SQS, DynamoDB | GitHub

- Built and deployed a CSV import service on **AWS** with CDK: 3 Lambda event source mappings, SQS, DynamoDB, and Cognito; all 16 live end-to-end checks passed.
- Made row writes safe to repeat with **DynamoDB** transactions: 9 live crash and duplicate tests saved no row twice, and fixes cut job time after a crash from 364.7 s to 4.8 s.

PUBLICATION

Coauthor, "Noninvasive Portable System With Dual Near-Infrared Light Technology for Monitoring Uric Acid Concentration," IEEE Transactions on Human-Machine Systems, accepted for publication, Aug 2026.